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## ASSIGNMENT -2

Subject : Computer Network

Subject Code : 24CSH-337

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**Question 1)** What is an IP Address? Why do we need classes in case of IPv4 address?

Explain each class, Mention how many host and networks are possible in each class.

Also explain what is subnetting and supernetting with examples.

**Answer)** An IP (Internet Protocol) address is a unique logical address assigned to a device connected to a network. It is used to identify the source and destination devices so that data packets can be delivered correctly. IPv4 is a classful addressing protocol. Its classes provide different size network block.

**Class A)** It's ranges from 0 – 127 in first octet with 0 and 127 being reserved addressed. There are over 16 Million hosts per network. There are 8 network bits and 24 host bits in Class A address.

**Class B)** It's ranges from 128 – 191 in first octet. There are over 65,000+ host per network. There are 16 network bits and 16 host bits.

**Class C)** its ranges from 192 – 223 in first octet. There are 254 hosts per network with 24 network bits and 8 host bits.

**Class D and Class E** are reserved for Multicast and Experiments respective.

Supernetting means combining multiple smaller networks into a larger network block. This helps reduce the number of routing table entries.

**Question 2)** Link-State v/s Distance-Vector Routing

**Answer)** link state and distance-vector routing algorithms are both used by routers to determine the least-cost path from a source to a destination.

Distance-vector uses Bellman – Ford algorithm to calculate the least-cost path, The routers have the distance with their neighbours. However, a sudden disconnect from a node may still be 'achievable' with infinite – updating.

Link-State uses Dijkstra algorithm to calculate the least-cost path. The routers in the network have the whole topology map, each router have same topology map.

**Question 3)** Congestion, Congestion Control, Leaky Bucket, Token Bucket and Fragmentation

**Answer)** Network congestion occurs when the amount of data entering a network is greater than the network's ability to process and forward it.

Congestion control is required to prevent excessive traffic from overwhelming network resources.

Without congestion control, packets may experience long delays, buffers may overflow, packets may be dropped, and retransmissions can further increase network traffic. This can significantly reduce network performance and may lead to congestion collapse.

Leaky Bucket algorithm controls traffic by transmitting packets a fixed rate. If more packets arrive when the bucket is full, the excess packets are discarded.

Token Bucket algorithm controls traffic by transmitting packets when enough tokens are generated. Tokens are generated at a fixed rate and store in a bucket.



